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1 Problem 1

1.1 Task

The p-q-formula of quadratic equations leads to $x^2 - 2px + q = 0$, \$x$_{1/2}$=p$\pm\sqrt{\{p^2 - q\}}$.\$$Writeafunction(p, q) \mapsto myPQnaive(p, q)$ that returns the zeros x_1 and x_2.

1.2 Notes

1.3 Solution

```
nil
```

2 Problem 2

2.1 Task

Visualize the following sequences (a) $a_n := (1 + \frac{1}{n})^n$ for $n = 1, 2, \ldots$ (b) Plot in the same graph: $\frac{1}{2^n}$ and $\frac{1}{n^2}$, for $n = 1, 2, \ldots$ (c) For $n \in \mathbb{N}$ put

$$b_n = \begin{cases} 1 & \text{if n is a square} \\ \dfrac{1}{n+1} & \text{otherwise} \end{cases}$$

(d) Find and illustrate the largest element in the sequence $(x_n)_{n\in\mathbb{N}}$, given by $x_n = \frac{n^2}{2^n}, \qquad n \in \mathbb{N}$

2.2 Notes

2.3 Solution

```
nil
```

3 Problem 3 DONE

3.1 Task

Show that (a) $\lim_{n\to\infty} \frac{(-1)^n}{n} = 0$, (b) $\lim_{n\to\infty} \frac{4n^2-2n+1}{n^2-n+6} = 4$ In addition, visualize this behavior in Julia.

3.2 Solution

- Part (a)

  We know that for all $n \geq 1$:

$$-1 \leq (-1)^n \leq 1$$

  Hence,

$$-\frac{1}{n} \leq \frac{(-1)^n}{n} \leq \frac{1}{n}$$

  Since, both $\lim_{n\to\infty} -\frac{1}{n} = 0$ and $\lim_{n\to\infty} \frac{1}{n} = 0$, $\lim_{n\to\infty} \frac{(-1)^n}{n} = 0$.

- Part (b)

  $\lim_{n\to\infty} \frac{4n^2-2n+1}{n^2-n+6} = 4$

$$\begin{aligned}
\lim_{n\to\infty} \frac{4n^2-2n+1}{n^2-n+6} &= \lim_{n\to\infty} \frac{4-\frac{2}{n}+\frac{1}{n^2}}{1-\frac{1}{n}+\frac{6}{n^2}} \quad \text{(divided by } n^2\text{)} \\
&= \frac{4-0+0}{1-0+0} \\
&= \frac{4}{1} \\
&= 4
\end{aligned}$$

- Code

```
using Plots

lim_1_term(n) = ((-1)^n)/(n)
lim_2_term(n) = (4n^2 - 2n + 1)/(n^2 - n + 6)

n = 1:100
lim_1 = lim_1_term.(n)
lim_2 = lim_2_term.(n)

p1 = scatter(n, lim_1,
             title="Limits",
             label="lim_1",
             xlabel="n", ylabel="term",
             legend=:bottomright)

p2 = scatter(n, lim_2,
             title="Limits",
```

```
                label="lim_2",
                xlabel="n", ylabel="term",
                legend=:bottomright)

  plot(p1, p2, layout=(2,1), size=(1200, 1200))

  savefig("./figs/p_03.png")
```

```
/Users/ziro2173/personal/university/01_bachelor/2025_winter_term/01_analysis/02_exercises/s
```

4 Problem 4

4.1 Task

Show that the sequence $(a_n)_{n\in\mathbb{N}}$, given by

$$a_n = n + n(-1)^n, \qquad n \in \mathbb{N}$$

is divergent. Check also in Julia.

4.2 Notes

4.3 Solution

```
nil
```

5 Problem 5

5.1 Task

Does the sequence $(x_n)_{n\in\mathbb{N}}$ given by

$$x_n = \sqrt{n+4} - \sqrt{n}, \qquad n \in \mathbb{N}$$

converge? Prove your answer and illustrate in Julia.

5.2 Notes

5.3 Solution

```
nil
```